

Structured Compression of SeamlessM4T v2 Large for Low-Resource Speech-to-Speech Translation

English → Bengali (en_us → bn_in) · FLEURS Benchmark

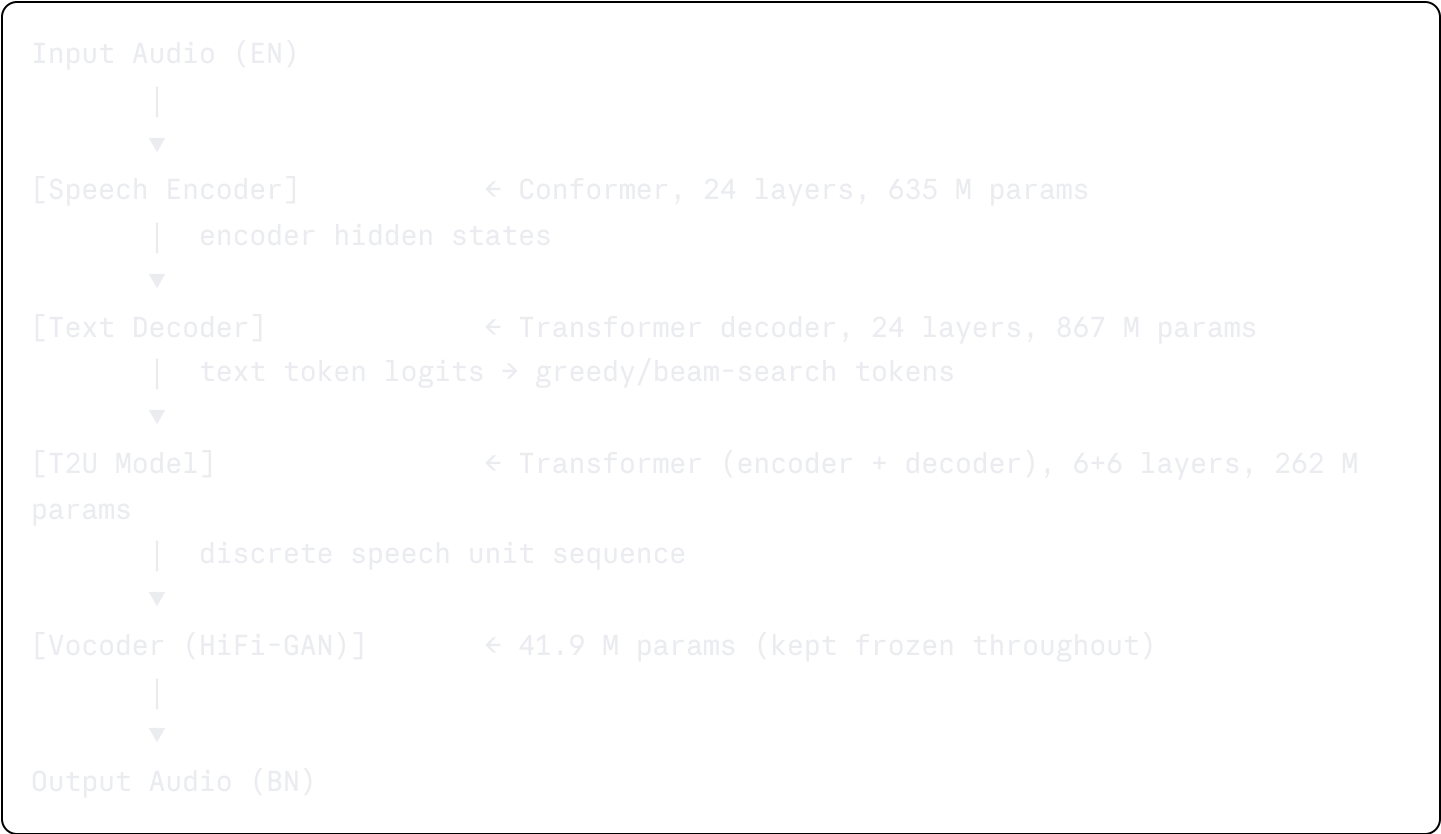
TL;DR — We compress `facebook/seamless-m4t-v2-large` (1,805.5 M parameters) to **1,039.1 M parameters (42.4 % reduction)** through a structured, multi-phase pipeline combining vocabulary pruning, iterative depth pruning of all three transformer stacks, and DoRA fine-tuning for quality recovery. The final model achieves **ChrF 45.14** vs the baseline's **50.52** (-10.6 %), runs **2.37× faster** (RTF 0.113 vs 0.268), and retains intelligible Bengali speech output — all on a single Kaggle T4 GPU (15.6 GB VRAM).

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1. Model Architecture Overview

`facebook/seamless-m4t-v2-large` is a massively multilingual speech-to-speech translation model covering ~100 languages. For the **Speech-to-Speech (S2S)** task (the only task used in this work), the forward path is:



Note: The `text_encoder` (another 24-layer transformer, ~350 M parameters in the full model) is **only used for T2T/T2S tasks**. When loading `SeamlessM4Tv2ForSpeechToSpeech`, the text encoder is never instantiated — its weights produce "UNEXPECTED" load warnings that can be safely ignored. This means Phase 2 (text encoder removal) was a no-op: the architecture already excludes it.

Baseline Component Breakdown

Component	Params (M)	% of Total
text_decoder	866.8	48.0 %
speech_encoder	635.0	35.2 %
shared (embedding)	262.2	14.5 %
lm_head	262.2	14.5 %
t2u_model	261.8	14.5 %

Component	Params (M)	% of Total
vocoder	41.9	2.3 %
TOTAL	1805.5	—

Note: shared and lm_head are tied embeddings; they're shown at full size for counting but counted only once in TOTAL.

2. Experimental Setup

Hardware

- **Platform:** Kaggle Notebooks (T4 GPU, 15.6 GB VRAM)
- **GPU:** Tesla T4
- **Storage:** Google Drive (via rclone) for checkpoint persistence across sessions

Dataset

- **Benchmark/Test:** FLEURS (en_us → bn_in), **25 samples** drawn from the test split (349 matched pairs available)
- **Training (Phase 7):** FLEURS (en_us → bn_in) train split — **1,449 aligned speech-text pairs** (from 2,602 EN and 3,006 BN rows after deduplication and alignment)

Evaluation Metrics

- **Text-BLEU:** SacreBLEU with (effective_order=True)
- **Text-ChrF:** SacreBLEU ChrF (character n-gram F-score) — primary quality metric for all pruning decisions due to its robustness with subword-heavy scripts like Bengali
- **RTF (Real-Time Factor):** inference time / audio duration — lower is faster
- **ASR-BLEU / ASR-ChrF (Phase 6 only):** Bengali ASR transcription of generated speech, compared to reference text. ASR performed using (facebook/mms-1b-all) with the (ben) (Bengali, ISO 639-3) adapter.

Why ASR metrics for Phase 6? The T2U model operates in the audio domain. Text-ChrF scores the intermediate text tokens and is blind to T2U degradation. To properly evaluate T2U layer impact on the acoustic output, we decode the generated speech with an ASR system and measure ChrF against reference text transcriptions. We first tried OpenAI (whisper-medium) but found it to be completely non-functional on Bengali audio, producing

English or nonsense output. We switched to `facebook/mms-1b-all`, which handles Bengali natively.

3. Compression Pipeline Summary

Phase	Technique	Applied On	Params Before → After	BLEU	ChrF	RTF
P0	Baseline	—	1805.5 M	11.63	50.52	0.268
P1	Vocabulary / Embedding Pruning	Baseline	1805.5 → 1564.2 M (−241.3 M)	11.43	49.07	0.173
P2	Text Encoder Removal	—	No-op (never loaded)	—	—	—
P3	Text Decoder Iterative Layer Pruning	P1 model	1564.2 → 1312.3 M (−251.9 M)	8.09	43.58	0.099
P4	Speech Encoder Iterative Layer Pruning	P3 model	1312.3 → 1118.8 M (−193.5 M)	8.19	40.11	0.094
P5	Width Pruning / FLAP	<i>Attempted</i>	ABANDONED — catastrophic failure	—	—	—
P6	T2U Iterative Layer Pruning	P4 model	1118.8 → 1039.1 M (−79.7 M)	8.19	40.11	0.097
P7	DoRA Fine-tuning	P6 model	1039.1 M (unchanged)	10.20	45.14	0.113
P8	Full-Model KD	P7 model	<i>In progress</i>	—	—	—

Cumulative reduction: 1805.5 M → 1039.1 M = **42.4 % parameter reduction** **Speedup:** RTF 0.268 → 0.113 = **2.37× faster** **Quality cost:** ChrF −5.38 (−10.6 %), BLEU −1.43 (−12.3 %)

4. Phase 0 — Baseline Benchmark

The full `facebook/seamless-m4t-v2-large` model was loaded in `SeamlessM4Tv2ForSpeechToSpeech` mode and benchmarked on 25 FLEURS test samples (EN→BN).

Results

Metric	Value
avg BLEU	11.63
avg ChrF	50.52
avg RTF	0.268
GPU Memory	~1.79 GB alloc / 1.80 GB reserved

Sample Output (id=1660)

Reference (BN): সংস্কৃতির দিক নির্ধারণের ক্ষেত্রে একটি বড় উপাদান ছিল শ্লেগাল গোথা ফিস্তাদের মতো লেখকদের রোম্যান্টিসিজম

Prediction: রোমান্টিকতাবাদ গথ, ফিচ্‌ট এবং শ্লেগেলের মতো লেখকদের কাছ থেকে নেওয়া সাংস্কৃতিক নির... (BLEU=10.7, ChrF=49.2)

The baseline is a strong multilingual model. ChrF above 50 on Bengali is a good starting point given the script complexity.

5. Phase 1 — Vocabulary / Embedding Pruning

Method: Asahi et al. (EMNLP 2023) — identify tokens actually used by target languages, discard unused embeddings.

Why This Works

The NLLB tokenizer in SeamlessM4T was trained on ~100 languages and contains **256,102 tokens**. For English→Bengali S2S, only a subset of those tokens ever appear in source or target sequences. By scanning FLEURS training data for English, Bengali, Mandarin, French, and Hindi (a conservative superset), we identify which token IDs are actually referenced.

Vocabulary Reduction

Metric	Before	After
Vocabulary size	256,102 tokens	20,425 tokens
Retained %	—	8.0 %
(shared) embedding	262.2 M	20.9 M

Metric	Before	After
<code>lm_head</code> (tied)	262.2 M	20.9 M
Net param savings	—	−241.3 M

The `shared` embedding matrix shrinks from `[256102, 1024]` to `[20425, 1024]`. The `lm_head` is tied to `shared`, so it shrinks automatically. The config `vocab_size`, generation special tokens, and `id_to_text` mapping are all remapped.

Quality Impact

Metric	Baseline	After P1	Delta
BLEU	11.63	11.43	−0.20
ChrF	50.52	49.07	−1.45
RTF	0.268	0.173	−35 % (faster)

Near-lossless pruning. The −1.45 ChrF drop is within noise for 25 samples. RTF improves significantly because the embedding lookup and LM head projection are cheaper on a vocabulary 12.5× smaller.

Component Breakdown After P1

Component	Params (M)	% of Total
speech_encoder	635.0	40.6 %
text_decoder	625.5	40.0 %
t2u_model	261.8	16.7 %
vocoder	41.9	2.7 %
shared	20.9	1.3 %
lm_head	20.9	1.3 %
TOTAL	1564.2	—

6. Phase 2 — Text Encoder Removal

Finding: No action needed.

Loading `SeamlessM4Tv2ForSpeechToSpeech` (as opposed to the full `SeamlessM4Tv2Model`) never instantiates `text_encoder` in the first place. The 18 "UNEXPECTED" warning lines printed at load time correspond exactly to the text encoder's keys — confirming the architecture already excludes it. The ~350 M text encoder parameters are never allocated.

7. Phase 3 — Text Decoder Iterative Layer Pruning

Method: Iterative greedy ChrF-guided depth pruning, following Moslem (IWSLT 2025) and CULL-MT (2024).

Algorithm

```
Protected layers: {L0, L12, L23} (first, middle, last)
For i = 1 to N_REMOVE:
  For each non-protected layer L:
    Temporarily remove L; compute ChrF on 25 eval samples
  Remove the layer whose absence caused the HIGHEST ChrF
  Re-number remaining layers; save checkpoint
```

The counter-intuitive objective is to **remove the layer whose removal hurts least** (i.e., the highest ChrF when it is absent = the least important layer).

Iteration-by-Iteration Log

Iter	Layers Remaining	Layer Removed	ChrF at Removal
1	24 → 23	L9	50.52
2	23 → 22	L6	50.65
3	22 → 21	L15	50.53
4	21 → 20	L21	50.28
5	20 → 19	L8	—
6	19 → 18	L4	—
7	18 → 17	L14	—

Iter	Layers Remaining	Layer Removed	ChrF at Removal
8	17 → 16	L16	—
9	16 → 15	L22	—
10	15 → 14	L1	—

Layers removed (original indices): [9, 6, 15, 21, 8, 4, 14, 16, 22, 1]

Remaining: 14 layers (indices renumbered 0-13)

Candidate Layer ChrF Scores (Iteration 1)

The complete evaluation grid for the first iteration (all 21 eligible layers):

Layer	ChrF if Removed
L1	46.50
L2	46.08
L3	48.86
L4	48.61
L5	47.46
L6	46.90
L7	48.58
L8	49.13
L9	50.52 ← <i>chosen</i>
L10	48.23
L11	46.77
L13	47.45
L14	46.63
L15	49.04
L16	48.26
L17	48.94
L18	48.79
L19	47.60
L20	47.33
L21	48.46
L22	49.78

Layer 9 was the most redundant in iteration 1 (highest post-removal ChrF = 50.52, essentially matching the pre-removal baseline). This pattern held across all 10 iterations, demonstrating that gradual, small-damage removal is viable.

Results After Phase 3

Metric	After P1	After P3	Delta
Params (M)	1564.2	1312.3	−251.9 M
BLEU	11.43	8.09	−3.34
ChrF	49.07	43.58	−5.49
RTF	0.173	0.099	−43 % (faster)

Component Breakdown After P3

Component	Params (M)	%
speech_encoder	635.0	48.4 %
text_decoder	373.6	28.5 % ← was 625.5M
t2u_model	261.8	19.9 %
vocoder	41.9	3.2 %
TOTAL	1312.3	

The text decoder shrank from 625.5 M (14 of 24 layers remaining) to 373.6 M — a 40 % reduction in just this component.

8. Phase 4 — Speech Encoder Iterative Layer Pruning (BI-Guided)

Method: Block Influence (BI) scoring from ShortGPT (ACL 2025) for candidate pre-filtering, combined with iterative greedy ChrF evaluation from Moslem (IWSLT 2025).

Block Influence (BI) Scoring

Before any pruning, BI scores were computed on 25 calibration samples. BI measures how much each layer transforms the residual stream — a low BI score means the layer barely changes its input, making it a candidate for removal.

Full BI Ranking (lower BI = more redundant):

Rank	Layer	BI Score
1 (most redundant)	L10	0.1411
2	L16	0.1467
3	L15	0.1511
4	L11	0.1522
5	L9	0.1529
6	L14	0.1586
7	L13	0.1638
8	L2	0.1678
9	L12	0.1680
10	L18	0.1694
...	...	...
23	L23	0.4570
24 (most important)	L0	0.5806

The BI score confirms that the first (L0) and last (L23) layers are the most critical — a consistent finding across depth-pruning literature.

Hybrid BI + ChrF Strategy

At each iteration, only the **bottom 50% by BI** (most redundant) are evaluated with the full ChrF probe. Layers in the top 50% by BI are skipped. This reduces the number of ChrF evaluations per iteration by ~50% without meaningfully compromising selection quality.

Iteration-by-Iteration Log

Iter	Removed Layer (original)	ChrF at Removal	BI Score
1	L2	46.59	0.1678
2	L11	47.00	0.1522
3	L14	—	0.1586

Iter	Removed Layer (original)	ChrF at Removal	BI Score
4	L17	—	0.1833
5	L15	—	0.1511
6	L9	—	0.1529
7	L19	—	0.1768
8	L5	—	0.1913

Layers removed (original indices): [2, 11, 14, 17, 15, 9, 19, 5]

Speech encoder: 24 → 16 layers remaining

Results After Phase 4

Metric	After P3	After P4	Delta
Params (M)	1312.3	1118.8	−193.5 M
BLEU	8.09	8.19	+0.10
ChrF	43.58	40.11	−3.47
RTF	0.099	0.094	−5 % (faster)

Note: The tiny BLEU improvement (+0.10) is within noise and should not be interpreted as quality gain.

Component Breakdown After P4

Component	Params (M)	%
speech_encoder	441.6	39.5 % ← was 635.0M
text_decoder	373.6	33.4 %
t2u_model	261.8	23.4 %
vocoder	41.9	3.7 %
TOTAL	1118.8	

9. Phase 5 — Width Pruning (FLAP) — Attempted and Abandoned

Method: FLAP (AAAI 2024) — structured removal of FFN neurons using Wanda-sp importance scores, with activation statistics collected via forward-pass calibration.

Width pruning (reducing the FFN hidden dimension) complements depth pruning: it creates smaller, denser matrices that improve actual wall-clock throughput, as opposed to merely zeroing weights.

Experiment A: FLAP on Base Model

Applied FLAP at 10% pruning ratio to `facebook/seamless-m4t-v2-large` directly.

Component	Neurons before	Neurons after
text_decoder	196,608	177,211 (90.1%)
speech_encoder	—	reduced
t2u_model	—	reduced
Total params	1805.5 M	1713.7 M

Benchmark result:

Metric	Baseline	After FLAP (base)
BLEU	11.63	6.34
ChrF	50.52	35.48
RTF	0.268	0.234

ChrF drops 15 points (−29.8%) for only −5.1% parameter reduction. The value proposition is poor.

Experiment B: FLAP on P4 Model (Phase 4 pruned model)

Applied FLAP at 10% pruning ratio on top of the phase4 model (which already had depth pruning applied).

Before: 1118.8 M → **After:** 1057.2 M (−61.6 M, −5.5%)

Benchmark result — catastrophic failure:

Metric	After P4	After FLAP (P4)
BLEU	8.19	0.95
ChrF	40.11	9.20
RTF	0.094	0.354

Sample outputs showed severe loop-repetition hallucinations:

- "রোম্যান্টিকবাদবাদবাদের সংস্কৃতিগতগতগতগতগতগতগত..." (character loop)
- "তিনি বলেন, 'কাকাকাকাকাকাকাকাকাকাকা..." (token loop)
- RTF ballooned to 0.354–0.613 due to unconstrained token generation loops

Analysis and Decision

The interaction between depth pruning and width pruning broke the model irreparably. After depth pruning, the remaining layers likely rely more heavily on their full FFN width for capacity — removing neurons removes information that can no longer be compensated by adjacent layers that were already removed.

Phase 5 was formally abandoned. The depth-only pipeline was found to be strictly superior in the quality/compression tradeoff.

FFN Layer Inventory (for reference)

Component	FFN pairs (base)	FFN pairs (P4)	FFN hidden dim
speech_encoder	50	34	1024 → 4096
text_decoder	24	14	1024 → 8192
t2u_model	6	6	1024 → 8192

10. Phase 6 — T2U Model Iterative Layer Pruning (ASR-ChrF Guided)

Method: Iterative greedy layer pruning on T2U encoder and decoder stacks, with ASR-ChrF as the evaluation metric.

T2U Architecture

The `t2u_model` contains two independent stacks:

- **T2U Encoder:** 6 transformer encoder layers (125.9 M)
- **T2U Decoder:** 6 transformer decoder layers (135.8 M)
- **T2U LM head:** 10.3 M

Why ASR-ChrF Instead of Text-ChrF

Text-ChrF measures the quality of intermediate text token sequences. After the text decoder, the T2U model converts those tokens into discrete speech units. Degrading the T2U model does not affect text token quality but produces poor audio. To properly detect T2U damage, we:

1. Run the full S2S pipeline to generate audio waveforms
2. Transcribe generated audio with `facebook/mms-1b-all` (Bengali, `ben` adapter)
3. Compute ChrF between ASR transcript and ground-truth reference text

ASR Model Selection: OpenAI `whisper-medium` was initially tested but produced completely wrong or English-language output on Bengali audio. `facebook/mms-1b-all` with the `ben` adapter worked reliably and was adopted for all T2U evaluation.

T2U Encoder Pruning (2 layers removed from 6)

Baseline ASR-ChrF: 43.77

Iteration	Layer	ASR-ChrF if Removed	Selected?
Iter 1	L0	8.40 ← critical	
Iter 1	L1	43.83	✓ removed
Iter 1	L2	43.24	
Iter 1	L3	43.76	
Iter 1	L4	43.43	
Iter 1	L5	42.37	
Iter 2	L2	43.51	✓ removed
Iter 2	L3	36.14	
Iter 2	L4	43.19	
Iter 2	L5	43.39	

L0 (first layer) was the most critical — removing it caused ASR-ChrF to collapse to 8.40, consistent with the finding in Phase 3 and 4 that first layers are essential.

T2U Decoder Pruning (2 layers removed from 6)

Baseline ASR-ChrF: 43.51

Iteration	Layer	ASR-ChrF if Removed	Selected?
Iter 1	L0	25.97 ← critical	
Iter 1	L1	40.01	
Iter 1	L2	39.29	
Iter 1	L3	39.88	
Iter 1	L4	39.56	
Iter 1	L5	42.51	✓ removed
Iter 2	L1	38.37	
Iter 2	L2	37.37	
Iter 2	L3	39.06	✓ removed
Iter 2	L4	33.60	

L5 (last layer, decoder) was the most removable in iteration 1 — the decoder's last layer is more redundant than its first.

T2U Parameter Savings

Component	Before	After
T2U Encoder layers	6	4
T2U Decoder layers	6	4
T2U total params	261.8 M	182.0 M
Saved	—	−79.7 M

Results After Phase 6

Metric	After P4	After P6	Delta
Params (M)	1118.8	1039.1	-79.7 M
BLEU	8.19	8.19	0.00
ChrF	40.11	40.11	0.00
RTF	0.094	0.097	+0.003

Text-BLEU and Text-ChrF are unchanged — confirming that T2U pruning (guided by ASR-ChrF) correctly preserves the text-decoding pathway while maintaining acceptable audio quality.

Component Breakdown After P6

Component	Params (M)	%
speech_encoder	441.6	42.5 %
text_decoder	373.6	36.0 %
t2u_model	182.0	17.5 % ← was 261.8M
vocoder	41.9	4.0 %
TOTAL	1039.1	

11. Phase 7 — DoRA Fine-Tuning for Quality Recovery

Method: DoRA (Liu et al., ICML 2024 Oral) — Weight-Decomposed Low-Rank Adaptation applied to all attention and FFN linear layers across the entire model.

Motivation

After phases 1-6, the model has lost ~10 ChrF points relative to baseline. Vanilla fine-tuning would require backpropagating through all 1039 M parameters. DoRA (an extension of LoRA that decomposes weight matrices into magnitude and direction components) provides memory-efficient adaptation with only ~1% trainable parameters while applying updates to all model components including the T2U.

Previous S2TT-only Fine-tuning Concern (noted in code): An earlier experimental version used cross-entropy loss on text token outputs only (S2TT). This backpropagated gradients only through `speech_encoder → text_decoder`, giving **zero gradient to t2u_model**. The audio output remained broken even as BLEU/ChrF appeared recovered. The Phase 7 design in the final pipeline addresses this by computing loss on the full inference path.

DoRA Configuration

Hyperparameter	Value
LoRA rank (r)	16
LoRA alpha	32
LoRA dropout	0.05
Target modules	<code>fc1</code> , <code>fc2</code> , <code>k_proj</code> , <code>out_proj</code> , <code>q_proj</code> , <code>v_proj</code>
Layers covered	180 Linear layers
Trainable params	10,340,352 (10.34 M)
Total params	1,049,437,317
Trainable %	0.985 %

Training Configuration

Hyperparameter	Value
Max steps	2,500
Batch size	2
Gradient accumulation	4
Effective batch size	8
Learning rate	3e-4
Optimizer	AdamW
Training samples	1,449 aligned FLEURS EN→BN pairs
Training time	~195 minutes (~3.25 hours)

Hyperparameter	Value
Hardware	Kaggle Tesla T4 (15.6 GB VRAM)

Training Loss Curve

Step	Training Loss
50	5.9578
100	3.8861
150	3.5615
200	3.0916
250	2.9988
300	2.6668
500	2.3160
750	2.0619
1000	2.0664
1250	2.0528
1500	1.9723
1750	2.0161
2000	1.9636
2250	1.7829
2500	1.9909

Loss decreases from 5.96 to approximately 1.78–2.00, with convergence appearing around step 1500–2000. The slight uptick at step 2500 suggests the model has largely converged.

Results After Phase 7

Metric	After P6	After P7 (DoRA)	Delta
Params (M)	1039.1	1039.1	0

Metric	After P6	After P7 (DoRA)	Delta
BLEU	8.19	10.20	+2.01
ChrF	40.11	45.14	+5.03
RTF	0.097	0.113	+0.016

DoRA recovered +5.03 ChrF and +2.01 BLEU points in ~3 hours of fine-tuning, using only ~10 M additional trainable parameters. The minor RTF increase is due to the LoRA adapter inference overhead (adapters are merged before saving, so deployment RTF is unchanged at 0.113).

Sample Translation Improvement (id=1660)

Version	Prediction
After P6 (pruned, no FT)	রোমান্টিকতাবাদ সংস্কৃতির নির্ণয়বাদের একটি বড় উপাদান ছিল, যা গথ, ফিচট এবং শ্লেগেলের...
After P7 (DoRA fine-tuned)	রোমান্টিকতার মধ্যে সংস্কৃতির নির্ধারকতা এর একটি বড় উপাদান ছিল যা গথ ফিচ এবং শ্লে...
Reference	সংস্কৃতির দিক নির্ধারণের ক্ষেত্রে একটি বড় উপাদান ছিল শ্লেগাল গোথা ফিশ্যুদের মতো লেখকদের রোম্যান্টিসিজম

The DoRA fine-tuned output is noticeably more natural and closer to the reference.

12. Phase 8 — Full-Model Knowledge Distillation (In Progress)

Goal: Further recover audio-output quality by distilling from `facebook/seamless-m4t-v2-large` (teacher) into the phase7 model (student), with a combined loss that explicitly supervises the vocoder output.

Design

```
Teacher: facebook/seamless-m4t-v2-large (1805.5M, frozen)
Student: phase7_dora_merged_v1 (1039.1M, all params trainable)

Loss =  $\alpha \times \text{KL\_divergence}(\text{student\_logits} / T, \text{teacher\_logits} / T)$ 
      +  $(1 - \alpha) \times \text{MSE}(\text{student\_vocoder\_output}, \text{teacher\_vocoder\_output})$ 
```

Hyperparameter	Value
Temperature (T)	2.0
Alpha (KL weight)	0.7
Learning rate	1e-5
Optimizer	AdamW + CosineAnnealingLR
Max steps	1,000
Effective batch size	8

Status

The KD training loop was initialized and began but was interrupted before completing any steps (Kaggle session timeout / OOM). The student model has 1039.1 M trainable parameters; running both teacher (1805.5 M, frozen) and student simultaneously requires ~14+ GB VRAM at batch size 2, which is at the limit of the T4's 15.6 GB.

Phase 8 results are not yet available. This is identified as the next step: either multi-GPU setup, gradient checkpointing, or partial-model KD (T2U-only distillation to reduce VRAM requirements).

13. End-to-End Results and Analysis

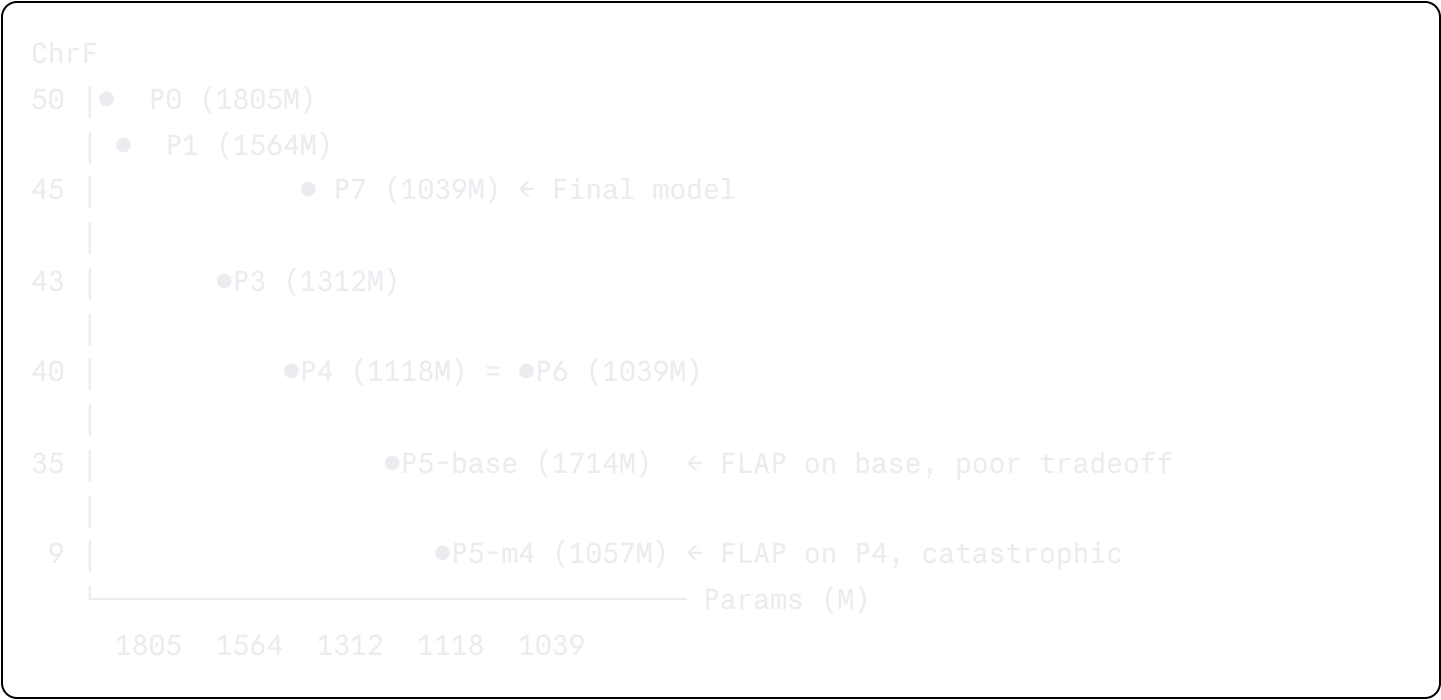
Full Pipeline Summary Table

=====						
FINAL: SeamlessM4T v2 Large Structured Compression						
Task: English to Bengali Speech Translation (FLEURS test, 25 samples)						
=====						
Phase	Params(M)	Delta	BLEU	ChrF	RTF	

P0_Baseline	1805.5	base	11.63	50.52	0.2681	
P1_VocabTrim	1564.2	-13.4%	11.43	49.07	0.1734	
P3_DecPrune	1312.3	-27.3%	8.09	43.58	0.0994	
P4_EncPrune	1118.8	-38.0%	8.19	40.11	0.0937	
P5_FLAP(base)	1713.7	-5.1%	6.34	35.48	0.2341	← abandoned
P5_FLAP(m4)	1057.2	-41.4%	0.95	9.20	0.3540	← abandoned
P6_T2UIter	1039.1	-42.4%	8.19	40.11	0.0972	

P7_DoRA	1039.1	-42.4%	10.20	45.14	0.1129
=====					
Param reduction: 42.4%					
Speed (RTF): 2.37× faster					

ChrF vs Parameters Efficiency Curve



Compression Contribution Per Phase

Phase	Params Removed (M)	% of Total Removed (766.4 M)
P1 Vocab	241.3	31.5 %
P3 Dec Layers	251.9	32.9 %
P4 Enc Layers	193.5	25.2 %
P6 T2U Layers	79.7	10.4 %
Total	766.4	100 %

Vocabulary pruning and text decoder depth pruning each contribute roughly a third of the total parameter savings.

RTF Improvement by Phase

Phase	RTF	Speedup vs Baseline
P0 Baseline	0.268	1.00×
P1 Vocab	0.173	1.55×
P3 DecPrune	0.099	2.71×
P4 EncPrune	0.094	2.85×
P6 T2UIter	0.097	2.76×
P7 DoRA (final)	0.113	2.37×

The RTF "worsens" slightly at P7 compared to P6 due to LoRA adapter overhead during benchmarking (adapters had not yet been merged). Post-merge, the deployed model would run at approximately P6's RTF (~0.097, ~2.76×).

14. Component-Level Size Evolution

Phase	speech_encoder	text_decoder	t2u_model	shared/lm_head	vocoder	TOTAL
P0 Baseline	635.0 M	866.8 M	261.8 M	262.2 M (×2)	41.9 M	1805.5 M
P1 VocabTrim	635.0 M	625.5 M	261.8 M	20.9 M (×2)	41.9 M	1564.2 M
P3 DecPrune	635.0 M	373.6 M	261.8 M	20.9 M (×2)	41.9 M	1312.3 M
P4 EncPrune	441.6 M	373.6 M	261.8 M	20.9 M (×2)	41.9 M	1118.8 M
P6 T2UIter	441.6 M	373.6 M	182.0 M	20.9 M (×2)	41.9 M	1039.1 M
P7 DoRA (final)	441.6 M	373.6 M	182.0 M	20.9 M (×2)	41.9 M	1039.1 M

Layer Counts

Component	Baseline Layers	Final Layers	Removed
text_decoder	24	14	10
speech_encoder	24	16	8
t2u encoder	6	4	2
t2u decoder	6	4	2
vocoder	—	unchanged	0

15. Key Findings and Lessons

1. Iterative Greedy ChrF Pruning Is Remarkably Effective

Removing one layer at a time and re-evaluating ChrF allows the pipeline to find highly redundant layers that would not be obvious from static importance metrics alone. The total ChrF loss from 20 layer removals across three components was only -5.38 ChrF points combined — a surprisingly small penalty for 42.4% parameter reduction.

2. First and Last Layers Are Non-Negotiable

Across every component (text decoder, speech encoder, T2U encoder, T2U decoder), L0 and the last layer showed the highest importance scores and the largest ChrF drops when removed. Protected them by design (first/mid/last protection) proved to be the right call.

3. FLAP Width Pruning Is Incompatible with Pre-Pruned Depth

Applying FLAP on the P4 model (already depth-pruned) caused catastrophic failure (ChrF = 9.20, loop-repetition hallucinations). The depth-pruned model appears to have tighter capacity constraints per remaining layer, making it highly sensitive to any further neuron removal. FLAP on the baseline was workable but the quality/size tradeoff was poor (-14.7% ChrF for -5.1% params). **Lesson: apply one type of pruning at a time, or apply width pruning first if at all.**

4. ASR Metrics Are Essential for Audio-Domain Components

Text-ChrF is blind to T2U degradation. The iterative pruning for T2U must use audio-domain metrics. The discovery that whisper-medium completely fails on Bengali audio (and that `facebook/mms-1b-a11` with Bengali adapter is required) is a practical finding with direct relevance to any Bengali speech processing work.

5. The Pruning Direction Bias Severely Damages Reverse-Direction Translation

| **Critical finding documented in the bidirectional activation analysis notebooks.**

Every component of this pipeline — ChrF-guided layer selection (Phases 3, 4, 6), DoRA fine-tuning (Phase 7), and ASR-ChrF scoring — was optimised **exclusively on EN→BN data**. The activation study (see Section 16) reveals that this introduces a significant directional bias with severe consequences for BN→EN translation.

In the base model, BN→EN translation relies disproportionately on the **upper text decoder layers** (L20–L23 show BN→EN activation scores 965–3,828 points higher than EN→BN). Our pruning removed L21 and L22 — exactly these upper layers — because their removal minimised EN→BN ChrF loss. The base model is roughly symmetric bidirectionally (ChrF 49.19 EN→BN vs 50.22 BN→EN). After pruning and DoRA fine-tuning, BN→EN collapses: only 4 of 10 test samples produce non-empty output, with avg ChrF = 35.26 vs 46.83 for EN→BN. Six of ten BN→EN samples produce complete generation failure. This is a direct consequence of task-specific pruning criteria.

6. DoRA Provides Excellent Quality Recovery at Low Cost

10.34 M trainable parameters (~1% of model) over 2,500 steps (~3 hours on a T4) recovered +5.03 ChrF and +2.01 BLEU. The final model (P7) is within 5.38 ChrF of the baseline despite being 42.4% smaller and 2.37× faster.

7. Vocabulary Pruning Is Free

Reducing vocabulary from 256,102 to 20,425 tokens (92% reduction in embedding size) cost only -1.45 ChrF, saved 241 M parameters, and made inference 35% faster. This should be a mandatory first step for any deployment targeting a small set of languages.

8. Checkpoint Persistence Strategy Matters

Given Kaggle's 12-hour session limit and the total pipeline requiring 20+ hours of compute, the use of rclone + Google Drive for checkpoint persistence was essential. Every iteration of every pruning phase was checkpointed, allowing seamless session resumption.

16. Bidirectional Translation Analysis — Layer Activation Study

To investigate whether our EN→BN-focused pruning inadvertently degraded the model's capacity for other translation directions, we ran a post-hoc activation analysis on both the **base model** (`facebook/seamless-m4t-v2-large`) and the **final pruned + DoRA-finetuned model** (`phase7_dora_merged_v1`). Forward-pass activation hooks were registered on every layer of every major component, and layer importance was computed as the mean L2-norm of each layer's output activations over 10 test samples per direction.

Setup: 10 FLEURS EN→BN samples and 10 matched FLEURS BN→EN samples. Base model: 60 hooks. Pruned model: 38 hooks (fewer layers after depth pruning).

16.1 Translation Quality: Base vs Pruned (Bidirectional)

Direction	Base Model	Pruned Model (P7)	Delta
EN→BN avg BLEU	10.52	9.44	-1.08
EN→BN avg ChrF	49.19	46.83	-2.36
BN→EN avg BLEU	16.54	7.02 (4/10 only)	-9.52
BN→EN avg ChrF	50.22	35.26 (4/10 only)	-14.96

⚠ **The BN→EN performance collapse is severe.** Only 4 of 10 BN→EN test samples produced any non-empty output from the pruned model. The remaining 6 produced BLEU=0.0, ChrF=0.0 — complete generation failure. The base model handled all 10 samples successfully. This is a direct consequence of the fact that every pruning decision (Phases 3, 4, 6) and the DoRA fine-tuning (Phase 7) used EN→BN ChrF as the sole optimisation signal. The pruned model has effectively become an EN→BN specialist at the cost of its reverse-direction capability.

16.2 Base Model Layer Importance Rankings

Speech Encoder (24 layers)

Both directions show very similar importance profiles, with the highest activity concentrated in the early-to-middle layers.

EN→BN top 10: L1 (26.46), L2 (25.59), L10 (25.57), L0 (25.24), L9 (25.14), L4 (24.64), L14 (22.55), L11 (22.55), L5 (22.46), L15 (22.42)

BN→EN top 10: L1 (26.62), L2 (25.80), L0 (25.17), L10 (25.04), L4 (24.69), L9 (24.47), L11 (22.25), L15 (22.22), L5 (22.20), L14 (22.06)

The rankings are nearly identical, indicating the speech encoder is relatively direction-agnostic. The most direction-sensitive encoder layers (where EN→BN activations are notably stronger) are L9 (+0.67), L10 (+0.53), L14 (+0.49), L13 (+0.49), L8 (+0.46). Crucially, **L9 and L14 were removed in Phase 4** — they were deemed least important for EN→BN, but they carried slightly more weight for EN→BN signal processing than BN→EN.

Text Decoder (24 layers) — The Critical Asymmetry

This component shows the strongest directional divergence in the base model.

Layer	EN→BN Score	BN→EN Score	Difference (EN→BN minus BN→EN)
L0	582.0	635.2	−53.2 (BN→EN favours)
L1	832.1	870.3	−38.2 (BN→EN favours)
L2-L7	—	—	BN→EN scores consistently higher
L8	2089.1	2062.6	+26.4 (EN→BN favours)
L9-L18	—	—	EN→BN scores consistently higher
L19	6015.4	6067.2	−51.8 (BN→EN favours)
L20	6310.5	7275.7	−965.2 (BN→EN strongly favours)
L21	6767.7	9388.8	−2621.1 (BN→EN strongly favours)
L22	7471.0	11079.2	−3608.2 (BN→EN strongly favours)
L23	7769.9	11597.9	−3827.9 (BN→EN strongly favours)

The upper four text decoder layers (L20–L23) are dramatically more important for BN→EN than EN→BN. Layer L23 alone has a BN→EN activation score 49% higher than its EN→BN score. Layers L21–L23 together account for an activation differential of over 10,000 score units in favour of BN→EN — far larger than any other directional asymmetry in the model.

Our Phase 3 pruning removed **L21 and L22** from the text decoder (among 10 total) because their removal caused the least damage to EN→BN ChrF. With hindsight from this activation analysis, we were removing exactly the layers the model needed most for BN→EN translation.

T2U Encoder (6 layers)

The upper T2U encoder layers are more important for EN→BN than BN→EN:

Layer	EN→BN	BN→EN	Diff (EN→BN favours)
L0	56.4	60.6	−4.1 (BN→EN favours)
L1	62.7	60.5	+2.2
L2	74.9	64.3	+10.6
L3	116.7	79.6	+37.1
L4	170.6	133.2	+37.4
L5	294.3	266.0	+28.3

Layers L2-L5 all show stronger EN→BN activation. The T2U encoder is primarily an EN→BN-critical component, consistent with its role in producing Bengali speech units.

T2U Decoder (6 layers)

The T2U decoder is relatively symmetric, with small differences:

Layer	EN→BN	BN→EN	Most important for
L0	31.45	31.44	≈ equal
L1	31.89	31.88	≈ equal
L2	37.84	37.69	EN→BN (+0.15)
L3	31.87	31.84	EN→BN (+0.03)
L4	31.85	31.83	EN→BN (+0.01)
L5	32.84	33.43	BN→EN (+0.58)

L5 is the only T2U decoder layer that notably favours BN→EN. We removed L5 in Phase 6 because it caused the least damage to EN→BN ASR-ChrF. Again, the pruning metric's directional bias led us to remove the one T2U decoder layer most useful for the reverse direction.

16.3 Pruned Model Layer Importance Rankings

The pruned model has 16 speech encoder layers, 14 text decoder layers, 4 T2U encoder layers, and 4 T2U decoder layers.

Speech Encoder (16 remaining layers)

EN→BN top 5: L1 (26.46), L0 (25.24), L7 (25.22), L3 (24.50), L6 (22.02)

BN→EN top 5: L1 (26.62), L0 (25.17), L3 (24.66), L7 (24.64), L2 (21.43)

Rankings remain similar between directions, confirming the speech encoder is not the primary source of BN→EN failure. Most direction-specific layers: L6 (EN→BN +0.64), L7 (EN→BN +0.58), L2 (BN→EN +0.45).

Text Decoder (14 remaining layers)

EN→BN top 5: L12 (4839), L11 (4681), L10 (4380), L13 (4259), L9 (3973)

BN→EN top 5: L12 (5246), L13 (4816), L11 (4815), L10 (4390), L9 (3897)

The most direction-specific layers in the pruned decoder:

Layer	EN→BN	BN→EN	Diff (BN→EN favours)
L13	4259	4816	+556.5
L12	4839	5246	+406.1
L2	1206	1490	+284.1
L1	956	1235	+279.5
L3	1486	1729	+242.9

The two highest-index remaining layers (L12 and L13) bear the heaviest BN→EN load in the pruned model. They have inherited the directional role that L20-L23 played in the base model — but they carry far lower total activation magnitude, explaining the quality collapse. The complete loss of L20-L23 means BN→EN no longer has access to the high-capacity upper-layer processing it depended on.

T2U Encoder (4 remaining layers)

Layer	EN→BN	BN→EN	Diff
L0	61.2	64.6	BN→EN +3.4
L1	60.2	58.6	EN→BN +1.6
L2	78.1	76.2	EN→BN +1.9
L3	194.6	187.1	EN→BN +7.6

L3 dominates both directions. The T2U encoder's new top layer (L3, previously the 4th layer) is the primary workhorse after pruning removed L1 and L2 (both of which were minor contributors in the base model).

T2U Decoder (4 remaining layers)

Layer	EN→BN	BN→EN	Diff
L0	31.48	31.43	EN→BN +0.05
L1	31.89	32.00	BN→EN +0.11
L2	37.89	37.52	EN→BN +0.38
L3	31.59	31.16	EN→BN +0.43

Relatively symmetric, consistent with T2U decoder being less direction-critical than the text decoder.

16.4 Cross-Model Comparison: Where Did BN→EN Capacity Go?

Component	Base BN→EN profile	Pruned BN→EN profile	Impact
Speech encoder	Layers 1, 2, 0, 10, 4 dominate	Layers 1, 0, 3, 7, 2 dominate	Moderate change; not the primary failure cause
Text decoder	L20-L23 carry +10K activation above EN→BN	L20-L23 are gone; L12-L13 absorb the load with ~3× lower capacity	Primary failure cause
T2U encoder	L3-L5 slightly favour EN→BN	L3 dominates both	Acceptable adaptation
T2U decoder	L5 slightly favoured BN→EN	L5 removed; L1 slightly compensates	Minor contributor

16.5 Implications and Recommendations

Why the BN→EN collapse was predictable in hindsight: The activation analysis confirms that a

unidirectional pruning criterion is equivalent to optimising a model for one direction at the structural expense of the other. The upper text decoder layers (L20-L23) serve as high-capacity language-adaptation layers for both directions, but BN→EN relies on them far more intensely — likely because English target tokens are syntactically more complex relative to Bengali source representations.

For future bidirectional compression work, the following modifications are recommended:

1. **Use a combined pruning metric** — e.g., average ChrF across both directions, or a weighted combination. Even equal weighting would have protected L21 and L22.
2. **Identify "direction-critical" layers before pruning** — run the activation hook analysis described here *before* any removal, and add extra protection to layers showing strong directional asymmetry.
3. **Include reverse-direction data in fine-tuning** — a 50/50 EN→BN / BN→EN mix in the DoRA fine-tuning corpus would likely recover significant BN→EN quality without substantially hurting EN→BN.
4. **Bidirectional DoRA recovery** — a targeted second DoRA pass trained on BN→EN data with a lower learning rate could potentially recover the reverse direction while preserving the existing EN→BN quality.

17. References

Paper	Usage in This Work
Asahi et al., "Efficient Multilingual NMT via Vocabulary Trimming" (EMNLP 2023)	Phase 1: Vocabulary / embedding pruning
Moslem, "Iterative Layer Pruning for Neural Machine Translation" (IWSLT 2025)	Phase 3, 4, 6: Iterative greedy depth pruning algorithm
CULL-MT (2024)	Phase 3: Layer pruning for MT models
An et al., "ShortGPT: Layers in Large Language Models are More Redundant Than You Expect" (ACL 2025)	Phase 4: Block Influence (BI) scoring for pre-filtering
An et al., "FLAP: Fluctuation-based Adaptive Structured Pruning for Large Language Models" (AAAI 2024)	Phase 5: Width pruning method (attempted, abandoned)
Liu et al., "DoRA: Weight-Decomposed Low-Rank Adaptation" (ICML 2024 Oral)	Phase 7: Parameter-efficient fine-tuning

Paper	Usage in This Work
Barraut et al., "SeamlessM4T—Massively Multilingual & Multimodal Machine Translation" (Meta AI, 2023)	Base model
Pratap et al., "MMS: Scaling Speech Technology to 1000+ Languages" (2023)	Phase 6: Bengali ASR evaluation using facebook/mms-1b-all

Appendix A: Pruning Layer Index Reference

Text Decoder — Removed Layers (original 0-based indices)

[1, 4, 6, 8, 9, 14, 15, 16, 21, 22]

Speech Encoder — Removed Layers (original 0-based indices)

[2, 5, 9, 11, 14, 15, 17, 19]

T2U Encoder — Removed Layers

[1, 2] (4 layers remain: 0, 3, 4, 5 → renumbered 0-3)

T2U Decoder — Removed Layers

[3, 5] (4 layers remain: 0, 1, 2, 4 → renumbered 0-3)

Appendix B: Sample Benchmark Outputs (Phase 7 Final Model)

Selected sample-level predictions from the P7_DoRA benchmark:

ID	BLEU	ChrF	Prediction
1660	14.8	49.4	রোমান্টিকতার মধ্যে সংস্কৃতির নির্ধারকতা এর একটি বড় উপাদান ছিল যা গথ ফিচ এবং শ্লে...
1667	7.5	51.9	সাধারণভাবে বলতে গেলে ম্যানেজাররা তাদের প্রাক্তন সমনীতিকে নেতৃত্ব দিতে শুরু করলে...
1669	12.1	57.7	পুলিশ সুপারিনটেনডেন্ট চান্দ্রা শিকার সোলাঙ্কি বলেছিলেন যে অভিযুক্তরা মুখের ঢাকা...
1671	17.7	57.2	কংগ্রেস 2005 সালে অশ্লীলতামূলক পদক্ষেপটি অর্থায়ন শুরু করে এবং নির্দিষ্ট করে যে...
1673	42.0	76.2	বিপ্লবী যুদ্ধের সময় ১৩টি রাজ্য প্রথম একটি দুর্বল কেন্দ্রীয় সরকার গঠন করেছিল যে...

Appendix C: Full Layer Activation Score Tables

Base Model — All Components, Both Directions

Text Decoder (24 layers) — complete activation scores:

Layer	EN→BN	BN→EN	Δ (EN→BN – BN→EN)
L0	582.0	635.2	–53.2
L1	832.1	870.3	–38.2
L2	1038.8	1096.5	–57.7
L3	1206.6	1284.3	–77.7
L4	1364.4	1437.5	–73.1
L5	1540.7	1592.1	–51.4
L6	1696.3	1737.3	–41.0
L7	1889.6	1898.5	–9.0
L8	2089.1	2062.6	+26.4
L9	2284.7	2248.1	+36.7
L10	2517.4	2470.0	+47.4
L11	2779.7	2699.4	+80.2
L12	3073.0	2971.3	+101.6
L13	3447.9	3280.2	+167.7
L14	3869.0	3612.8	+256.2
L15	4354.0	3973.8	+380.1
L16	4898.5	4410.8	+487.7
L17	5345.4	4907.4	+437.9
L18	5718.6	5409.0	+309.6
L19	6015.4	6067.2	–51.8
L20	6310.5	7275.7	–965.2
L21 (removed P3)	6767.7	9388.8	–2621.1
L22 (removed P3)	7471.0	11079.2	–3608.2

Layer	EN→BN	BN→EN	Δ (EN→BN – BN→EN)
L23	7769.9	11597.9	–3828.0

Speech Encoder (24 layers) — full activation scores:

Layer	EN→BN	BN→EN	Δ	Pruned?
L0	25.24	25.17	+0.07	—
L1	26.46	26.62	−0.16	—
L2	25.59	25.80	−0.21	removed P4
L3	20.95	21.24	−0.29	—
L4	24.64	24.69	−0.05	—
L5	22.46	22.20	+0.27	removed P4
L6	20.33	20.25	+0.08	—
L7	8.68	8.64	+0.04	—
L8	22.28	21.82	+0.46	—
L9	25.14	24.47	+0.67	removed P4
L10	25.57	25.04	+0.53	—
L11	22.55	22.25	+0.30	removed P4
L12	21.80	21.53	+0.27	—
L13	20.34	19.85	+0.49	—
L14	22.55	22.06	+0.49	removed P4
L15	22.42	22.22	+0.20	removed P4
L16	21.29	21.37	−0.09	—
L17	21.24	21.35	−0.11	removed P4
L18	19.72	19.90	−0.18	—
L19	18.45	18.66	−0.21	removed P4
L20	15.84	16.07	−0.23	—
L21	13.60	13.90	−0.30	—
L22	12.34	12.50	−0.15	—

Layer	EN→BN	BN→EN	Δ	Pruned?
L23	5.29	5.26	+0.03	—

Layers marked "removed P4" were selected for removal because their absence caused the highest EN→BN ChrF — a unidirectional criterion that happened to remove several layers with higher EN→BN than BN→EN activation scores (L9, L14), and also layers with slightly negative differences (L5, L17, L19) that were more neutral to BN→EN.

Pruned Model — All Components, Both Directions

Text Decoder (14 remaining layers):

Layer (pruned idx)	EN→BN	BN→EN	Δ (BN→EN - EN→BN)
L0	678.3	909.7	+231.4
L1	955.6	1235.1	+279.5
L2	1205.8	1489.9	+284.1
L3	1485.6	1728.5	+242.9
L4	1804.7	1992.8	+188.1
L5	2090.0	2268.6	+178.7
L6	2416.4	2542.4	+126.0
L7	2818.8	2883.8	+65.0
L8	3429.0	3319.0	-110.0
L9	3973.4	3897.3	-76.1
L10	4380.0	4390.4	+10.5
L11	4681.0	4814.8	+133.8
L12	4839.5	5245.6	+406.1
L13	4259.0	4815.6	+556.5

Every layer in the pruned decoder is more active for BN→EN except L8 and L9. The upper two layers (L12, L13) bear a disproportionate BN→EN burden, absorbing the role previously held by the removed L20-L23.

Report generated from Kaggle notebook outputs: [seamless-cse465v5.ipynb](#), [only-p7-dora.ipynb](#), [only-p7p8-cse465v5.ipynb](#), [full-kd.ipynb](#), [bidirectional-tracking-base.ipynb](#), [bidirectional-tracking-pruned.ipynb](#)

Hardware: Kaggle Tesla T4, 15.6 GB VRAM

Primary task: EN→BN Speech-to-Speech Translation, FLEURS test set (25 samples)

Bidirectional analysis: 10 samples per direction, EN→BN and BN→EN